

✓ LAB 3

Control Flow for Data Cleaning

UCI Heart Disease Dataset

```
url <- "https://archive.ics.uci.edu/ml/machine-learning-databases/heart-disease  
  
heart <- read.csv(  
  url,  
  header = FALSE,  
  na.strings = "?"  
)
```

```
colnames(heart) <- c(  
  "age",  
  "sex",  
  "cp",  
  "trestbps",  
  "chol",  
  "fbs",  
  "restecg",  
  "thalach",  
  "exang",  
  "oldpeak",  
  "slope",  
  "ca",  
  "thal",  
  "target"  
)
```

```
head(heart)
```

A data.frame: 6 × 14										
	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	63	1	1	145	233	1	2	150	0	2.3
2	67	1	4	160	286	0	2	108	1	1.5
3	67	1	4	120	229	0	2	129	1	2.6
4	37	1	3	130	250	0	0	187	0	3.5
5	41	0	2	130	204	0	2	172	0	1.4
6	56	1	2	120	236	0	0	178	0	0.8

summary(heart)			
age	sex	cp	trestbps
Min. :29.00	Min. :0.0000	Min. :1.000	Min. : 94.0
1st Qu.:48.00	1st Qu.:0.0000	1st Qu.:3.000	1st Qu.:120.0
Median :56.00	Median :1.0000	Median :3.000	Median :130.0
Mean :54.44	Mean :0.6799	Mean :3.158	Mean :131.7
3rd Qu.:61.00	3rd Qu.:1.0000	3rd Qu.:4.000	3rd Qu.:140.0
Max. :77.00	Max. :1.0000	Max. :4.000	Max. :200.0
chol	fbs	restecg	thalach
Min. :126.0	Min. :0.0000	Min. :0.0000	Min. : 71.0
1st Qu.:211.0	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:133.5
Median :241.0	Median :0.0000	Median :1.0000	Median :153.0
Mean :246.7	Mean :0.1485	Mean :0.9901	Mean :149.6
3rd Qu.:275.0	3rd Qu.:0.0000	3rd Qu.:2.0000	3rd Qu.:166.0
Max. :564.0	Max. :1.0000	Max. :2.0000	Max. :202.0
exang	oldpeak	slope	ca
Min. :0.0000	Min. :0.00	Min. :1.000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.00	1st Qu.:1.000	1st Qu.:0.0000
Median :0.0000	Median :0.80	Median :2.000	Median :0.0000
Mean :0.3267	Mean :1.04	Mean :1.601	Mean :0.6722
3rd Qu.:1.0000	3rd Qu.:1.60	3rd Qu.:2.000	3rd Qu.:1.0000
Max. :1.0000	Max. :6.20	Max. :3.000	Max. :3.0000
			NAs :4
thal	target		
Min. :3.000	Min. :0.0000		
1st Qu.:3.000	1st Qu.:0.0000		
Median :3.000	Median :0.0000		
Mean :4.734	Mean :0.9373		
3rd Qu.:7.000	3rd Qu.:2.0000		
Max. :7.000	Max. :4.0000		
NAs :2			

```
names(heart)
```

```
'age' · 'sex' · 'cp' · 'trestbps' · 'chol' · 'fbs' · 'restecg' · 'thalach' · 'exang' · 'oldpeak' · 'slope' · 'ca' ·  
'thal' · 'target'
```

```
summary(heart$trestbps)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
94.0	120.0	130.0	131.7	140.0	200.0

```
head(heart$trestbps, 10)
```

```
145 · 160 · 120 · 130 · 130 · 120 · 140 · 120 · 130 · 140
```

```
heart_clean <- heart
```

```
set.seed(123)
```

```
heart_clean$trestbps[1] <- -120  
heart_clean$trestbps[2] <- NA  
heart_clean$trestbps[3] <- 320  
heart_clean$trestbps[4] <- 350
```

```
heart_clean$trestbps[1:10]
```

```
-120 · <NA> · 320 · 350 · 130 · 120 · 140 · 120 · 130 · 140
```

```
clean_bp <- function(bp) {  
  
  if (is.na(bp)) {  
    return(NA)  
  } else if (bp < 0) {  
    return(NA)  
  } else if (bp > 250) {  
    return(250)  
  } else {  
    return(bp)  
  }  
}
```

```
heart_clean$clean_trestbps <- sapply(  
  heart_clean$trestbps,  
  clean_bp  
)
```

```
heart_clean$clean_trestbps[1:10]
```

<NA> · <NA> · 250 · 250 · 130 · 120 · 140 · 120 · 130 · 140

```
safe_mean <- function(x) {  
  
  tryCatch(  
    {  
      if (all(is.na(x))) {  
        stop("Cannot calculate mean: all values are missing.")  
      }  
  
      mean(x, na.rm = TRUE)  
    },  
  
    warning = function(w) {  
      message("Warning: Missing values detected. Calculating mean using available values.")  
      mean(x, na.rm = TRUE)  
    },  
  
    error = function(e) {  
      message("Error: ", e$message)  
      return(NA)  
    }  
  )  
}
```

```
safe_mean(heart_clean$clean_trestbps)
```

132.382059800664

```
safe_ratio <- function(chol, bp) {  
  
  tryCatch(  
    {  
      if (is.na(bp)) {  
        stop("Invalid denominator: BP is NA.")  
      }  
  
      if (bp == 0) {  
        stop("Invalid denominator: BP is zero.")  
      }  
  
      if (bp < 0 || bp > 250) {  
        stop("Invalid denominator: BP is outside valid range.")  
      }  
  
      chol / bp  
    },  
  
    error = function(e) {  
      message("Error: ", e$message)  
      return(NA)  
    }  
  )  
}
```

```

        message("Ratio Error: ", e$message)
        return(NA)
    }
)
}

```

```

safe_ratio(
  heart_clean$chol[1],
  heart_clean$clean_trestbps[1]
)

```

Ratio Error: Invalid denominator: BP is NA.

<NA>

```

safe_ratio(
  heart_clean$chol[5],
  heart_clean$clean_trestbps[5]
)

```

1.56923076923077

```

loop_clean <- function(x) {

  result <- x

  for (i in seq_along(x)) {

    if (is.na(x[i])) {
      result[i] <- NA
    } else if (x[i] < 0) {
      result[i] <- NA
    } else if (x[i] > 250) {
      result[i] <- 250
    }
  }

  return(result)
}

```

```

loop_result <- loop_clean(heart_clean$trestbps)

```

```

loop_result[1:10]

```

<NA> · <NA> · 250 · 250 · 130 · 120 · 140 · 120 · 130 · 140

```

vector_clean <- function(x) {

```

```

result <- x

result[result < 0] <- NA
result[result > 250] <- 250

return(result)
}

```

```

vector_result <- vector_clean(heart_clean$trestbps)

vector_result[1:10]

```

```

<NA> · <NA> · 250 · 250 · 130 · 120 · 140 · 120 · 130 · 140

```

```

all.equal(
  loop_result,
  vector_result,
  check.attributes = FALSE
)

```

```

TRUE

```

```

loop_time <- system.time({
  loop_result <- loop_clean(heart_clean$trestbps)
})

```

```

loop_time

```

```

user  system elapsed
0      0      0

```

```

vector_time <- system.time({
  vector_result <- vector_clean(heart_clean$trestbps)
})

```

```

vector_time

```

```

user  system elapsed
0.003  0.000  0.003

```

```

comparison <- data.frame(
  Method = c("For Loop", "Vectorized"),
  User_Time = c(
    loop_time["user.self"],
    vector_time["user.self"]
  ),
  System_Time = c(
    loop_time["sys.self"],

```

```

        vector_time["sys.self"]
      ),
      Elapsed_Time = c(
        loop_time["elapsed"],
        vector_time["elapsed"]
      )
    )
  )

comparison

```

A data.frame: 2 × 4

Method	User_Time	System_Time	Elapsed_Time
<chr>	<dbl>	<dbl>	<dbl>
For Loop	0.000	0	0.000
Vectorized	0.003	0	0.003

✓ Loop vs Vectorized Cleaning — Conclusion

Both the for-loop and vectorized approaches successfully cleaned the invalid blood-pressure values and produced the same results.

The execution-time comparison shows the performance difference between the two approaches. The vectorized approach is generally more efficient for operations on complete R vectors, while the for-loop processes observations individually.

```

cat("Missing BP values:", sum(is.na(heart_clean$clean_trestbps)), "\n")

cat("Minimum BP:", min(heart_clean$clean_trestbps, na.rm = TRUE), "\n")

cat("Maximum BP:", max(heart_clean$clean_trestbps, na.rm = TRUE), "\n")

cat("Mean BP:", mean(heart_clean$clean_trestbps, na.rm = TRUE), "\n")

cat("Median BP:", median(heart_clean$clean_trestbps, na.rm = TRUE), "\n")

cat("Any negative BP values:",
    any(heart_clean$clean_trestbps < 0, na.rm = TRUE), "\n")

cat("Any BP values greater than 250:",
    any(heart_clean$clean_trestbps > 250, na.rm = TRUE), "\n")

```

```

Missing BP values: 2
Minimum BP: 94
Maximum BP: 250
Mean BP: 132.3821
Median BP: 130

```

```
Any negative BP values: FALSE
Any BP values greater than 250: FALSE
```

```
write.csv(
  heart_clean,
  "cleaned_heart_data.csv",
  row.names = FALSE
)
```

```
file.exists("cleaned_heart_data.csv")
```

```
TRUE
```

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The BP cleaning function successfully handled negative, missing, and extreme values. Both loop and vectorized methods produced correct results. The vectorized method was generally more efficient than the loop-based method.

The BP cleaning function successfully handled negative, missing, and extreme values. Both loop and vectorized methods produced correct results. The vectorized method was generally more efficient than the loop-based method.

✓ LAB 4

Advanced Missing Data Handling

NA, NULL, NaN, Missing-Value Detection and Imputation

UCI Adult / Census Income Dataset

```
install.packages("naniar")
install.packages("skimr")
```

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

also installing the dependencies ‘gridExtra’, ‘plyr’, ‘norm’, ‘visdat’, ‘viridis’

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)


```
library(naniar)
library(skimr)
```

Attaching package: 'skimr'

The following object is masked from 'package:naniar':

```
n_complete
```

```
url_adult <- "https://archive.ics.uci.edu/ml/machine-learning-databases/adult/æ

adult <- read.csv(
  url_adult,
  header = FALSE,
  na.strings = "?",
  strip.white = TRUE
)
```

```
colnames(adult) <- c(
  "age",
  "workclass",
  "fnlwgt",
  "education",
  "education_num",
  "marital_status",
  "occupation",
  "relationship",
  "race",
  "sex",
  "capital_gain",
  "capital_loss",
  "hours_per_week",
  "native_country",
  "income"
)
```

```
head(adult)
```

	age	workclass	fnlwgt	education	education_num	marital_status	occupatio
	<int>	<chr>	<int>	<chr>	<int>	<chr>	<chr>
1	39	State-gov	77516	Bachelors	13	Never-married	Adm clerical
2	50	Self-emp- not-inc	83311	Bachelors	13	Married-civ- spouse	Exec managerial
3	38	Private	215646	HS-grad	9	Divorced	Handlers cleaner
4	53	Private	234721	11th	7	Married-civ- spouse	Handlers cleaner
5	28	Private	338409	Bachelors	13	Married-civ- spouse	Pro specialty
6	37	Private	284582	Masters	14	Married-civ- spouse	Exec managerial

```
str(adult)
```

```
'data.frame':  32561 obs. of  15 variables:
 $ age          : int  39 50 38 53 28 37 49 52 31 42 ...
 $ workclass    : chr  "State-gov" "Self-emp-not-inc" "Private" "Private" ...
 $ fnlwgt       : int  77516 83311 215646 234721 338409 284582 160187 209642 45
 $ education    : chr  "Bachelors" "Bachelors" "HS-grad" "11th" ...
 $ education_num : int  13 13 9 7 13 14 5 9 14 13 ...
 $ marital_status: chr  "Never-married" "Married-civ-spouse" "Divorced" "Married
 $ occupation   : chr  "Adm-clerical" "Exec-managerial" "Handlers-cleaners" "Ha
 $ relationship : chr  "Not-in-family" "Husband" "Not-in-family" "Husband" ...
 $ race         : chr  "White" "White" "White" "Black" ...
 $ sex          : chr  "Male" "Male" "Male" "Male" ...
 $ capital_gain : int  2174 0 0 0 0 0 0 0 14084 5178 ...
 $ capital_loss : int  0 0 0 0 0 0 0 0 0 0 ...
 $ hours_per_week: int  40 13 40 40 40 40 16 45 50 40 ...
 $ native_country: chr  "United-States" "United-States" "United-States" "United-
 $ income       : chr  "<=50K" "<=50K" "<=50K" "<=50K" ...
```

```
summary(adult)
```

age	workclass	fnlwgt	education
Min. :17.00	Length :32561	Min. : 12285	Length :32561
1st Qu.:28.00	N.unique : 8	1st Qu.: 117827	N.unique : 16
Median :37.00	N.blank : 0	Median : 178356	N.blank : 0
Mean :38.58	Min.nchar: 7	Mean : 189778	Min.nchar: 3
3rd Qu.:48.00	Max.nchar: 16	3rd Qu.: 237051	Max.nchar: 12
Max. :90.00	NAs : 1836	Max. :1484705	
education_num	marital_status	occupation	relationship
Min. : 1.00	Length :32561	Length :32561	Length :32561
1st Qu.: 9.00	N.unique : 7	N.unique : 14	N.unique : 6
Median :10.00	N.blank : 0	N.blank : 0	N.blank : 0
Mean :10.08	Min.nchar: 7	Min.nchar: 5	Min.nchar: 4
3rd Qu.:12.00	Max.nchar: 21	Max.nchar: 17	Max.nchar: 14
Max. :16.00		NAs : 1843	
race	sex	capital_gain	capital_loss
Length :32561	Length :32561	Min. : 0	Min. : 0.0
N.unique : 5	N.unique : 2	1st Qu.: 0	1st Qu.: 0.0
N.blank : 0	N.blank : 0	Median : 0	Median : 0.0
Min.nchar: 5	Min.nchar: 4	Mean : 1078	Mean : 87.3
Max.nchar: 18	Max.nchar: 6	3rd Qu.: 0	3rd Qu.: 0.0
		Max. :99999	Max. :4356.0
hours_per_week	native_country	income	
Min. : 1.00	Length :32561	Length :32561	
1st Qu.:40.00	N.unique : 41	N.unique : 2	
Median :40.00	N.blank : 0	N.blank : 0	
Mean :40.44	Min.nchar: 4	Min.nchar: 4	
3rd Qu.:45.00	Max.nchar: 26	Max.nchar: 5	
Max. :99.00	NAs : 583		

```
dim(adult)
```

```
32561 · 15
```

```
names(adult)
```

```
'age' · 'workclass' · 'fnlwgt' · 'education' · 'education_num' · 'marital_status' · 'occupation' ·  
'relationship' · 'race' · 'sex' · 'capital_gain' · 'capital_loss' · 'hours_per_week' · 'native_country' ·  
'income'
```

```
adult_clean <- adult
```

```
adult_clean$age[1] <- NA
```

```
adult_clean$age[2] <- NaN
```

```
adult_clean$workclass[3] <- ""
```

```
adult_clean$age[4] <- 999
```

```
cat("First 5 ages:\n")
print(adult_clean$age[1:5])

cat("\nFirst 5 workclass values:\n")
print(adult_clean$workclass[1:5])
```

```
First 5 ages:
[1] NA NaN 38 999 28

First 5 workclass values:
[1] "State-gov"          "Self-emp-not-inc" ""          "Private"
[5] "Private"
```

```
is.na(adult_clean$age[1])
```

```
TRUE
```

```
x <- NULL
```

```
is.null(x)
```

```
TRUE
```

✓ Difference between NA, NaN, NULL and ""

- NA represents missing data.
- NaN represents an undefined numerical value.
- NULL represents the absence of an R object.
- "" represents an empty character string.

NULL normally does not behave like a missing value inside a data-frame cell.

```
adult_clean$workclass[3]
```

```
"
```

```
adult_clean$workclass[3] == ""
```

```
TRUE
```

```
sum(!is.na(adult_clean$workclass) & adult_clean$workclass == "")
```

```
1
```

```
miss_var_summary(adult_clean)
```

A tibble: 15 × 3

variable	n_miss	pct_miss
<chr>	<int>	<num>
occupation	1843	5.66
workclass	1836	5.64
native_country	583	1.79
age	2	0.006 [4m1 [24m [4m4 [24m
fnlwgt	0	0
education	0	0
education_num	0	0
marital_status	0	0
relationship	0	0
race	0	0
sex	0	0
capital_gain	0	0
capital_loss	0	0
hours_per_week	0	0
income	0	0

```
missing_before <- colSums(is.na(adult_clean))
```

```
missing_before
```

```
age:      2 workclass:      1836 fnlwgt:      0 education:      0 education_num:      0
marital_status:      0 occupation:      1843 relationship:      0 race:      0 sex:      0
capital_gain:      0 capital_loss:      0 hours_per_week:      0 native_country:      583
income:      0
```

```
adult_clean$age[adult_clean$age == 999] <- NA
```

```
sum(adult_clean$age == 999, na.rm = TRUE)
```

```
1
```

```
adult_clean$workclass[
  !is.na(adult_clean$workclass) & adult_clean$workclass == ""
] <- "Unknown"
```

```
sum(
  !is.na(adult_clean$workclass) &
  adult_clean$workclass == ""
)
```

0

```
median_impute <- function(x) {

  median_value <- median(x, na.rm = TRUE)

  x[is.na(x)] <- median_value

  return(x)
}
```

```
numeric_columns <- c(
  "age",
  "fnlwgt",
  "education_num",
  "capital_gain",
  "capital_loss",
  "hours_per_week"
)
```

```
for (col in numeric_columns) {
  adult_clean[[col]] <- median_impute(adult_clean[[col]])
}
```

```
missing_after <- colSums(is.na(adult_clean))
```

missing_after

age:	0	workclass:	1836	fnlwgt:	0	education:	0	education_num:	0
marital_status:	0	occupation:	1843	relationship:	0	race:	0	sex:	0
capital_gain:	0	capital_loss:	0	hours_per_week:	0	native_country:	583		
income:	0								

```
comparison_missing <- data.frame(
  Variable = names(missing_before),
  Before = as.numeric(missing_before),
  After = as.numeric(missing_after)
)
```

comparison_missing

A data.frame: 15 × 3

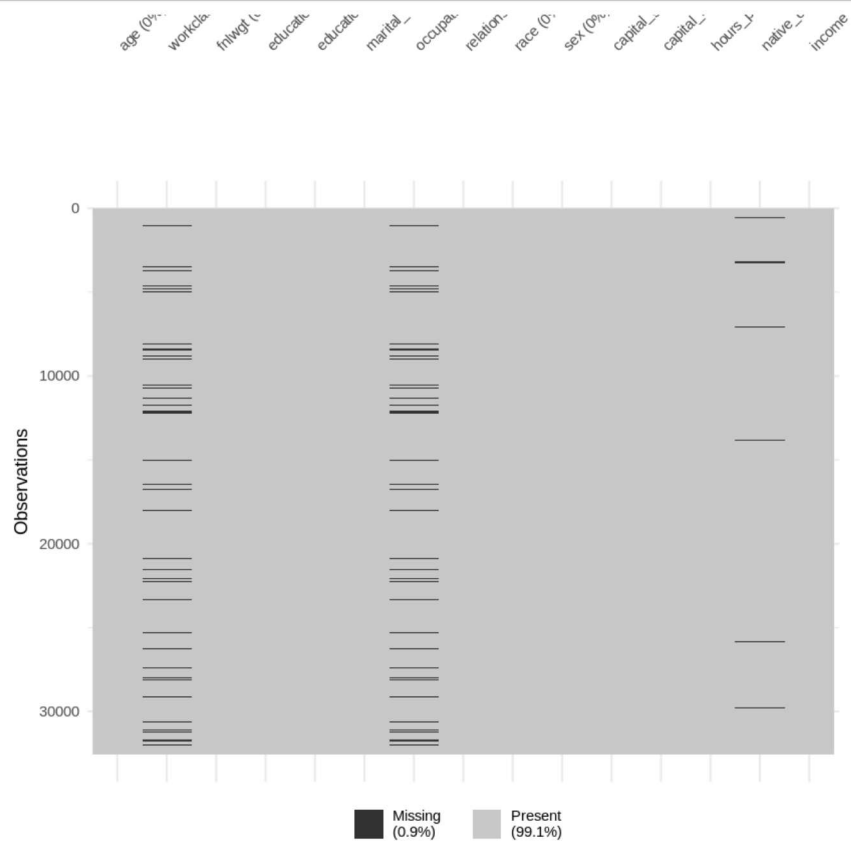
Variable	Before	After
<chr>	<dbl>	<dbl>
age	2	0
workclass	1836	1836
fnlwgt	0	0
education	0	0
education_num	0	0
marital_status	0	0
occupation	1843	1843
relationship	0	0
race	0	0
sex	0	0
capital_gain	0	0
capital_loss	0	0
hours_per_week	0	0
native_country	583	583
income	0	0

```
complete_rows <- complete.cases(adult_clean)
```

```
table(complete_rows)
```

```
complete_rows  
FALSE TRUE  
2399 30162
```

```
vis_miss(adult_clean)
```



```
skim(adult_clean)
```


	skim_type	skim_variable	n_missing	complete_rate	character.min	character
	<chr>	<chr>	<int>	<dbl>	<int>	.
1	character	workclass	1836	0.9436135	7	.

```
cat(
  "Age = 999 remaining:",
  any(adult_clean$age == 999, na.rm = TRUE),
  "\n"
)
```

```
cat(
  "Blank workclass remaining:",
  any(
    !is.na(adult_clean$workclass) &
    adult_clean$workclass == ""
  ),
  "\n"
)
```